

AI-Based Microplastic Contamination Risk Assessment System Using Light Scattering and Multi-Angle Detection

Abstract

Microplastic pollution has emerged as a significant environmental concern due to its widespread presence in rivers, lakes, oceans, and drinking water sources. Traditional laboratory-based microplastic detection methods are expensive, time-consuming, and require specialized equipment. This project proposes a low-cost IoT-based system for estimating microplastic contamination risk in water using optical sensing principles, including light scattering, multi-angle detection, and turbidity analysis. The system utilizes an ESP32 microcontroller, optical sensors, LEDs, a turbidity sensor, and a floating sampling platform to continuously monitor water quality. Machine learning techniques are incorporated to classify contamination levels into different risk categories. The proposed system aims to provide an affordable and scalable solution for environmental monitoring and early contamination assessment.

Chapter 1: Introduction

1.1 Background

Microplastics are plastic particles smaller than 5 mm that originate from the degradation of larger plastic materials or industrial processes. Due to their small size, they can enter aquatic ecosystems and potentially affect both environmental and human health.

Conventional microplastic detection methods typically involve laboratory procedures such as spectroscopy, microscopy, and chemical analysis. While highly accurate, these techniques require expensive instruments and trained personnel.

Recent advancements in IoT and embedded systems have enabled the development of low-cost monitoring solutions. This project explores the possibility of estimating microplastic contamination risk using optical properties of water samples.

1.2 Problem Statement

Existing microplastic detection techniques are often expensive and unsuitable for continuous field deployment. There is a need for an affordable system capable of monitoring water quality and estimating microplastic contamination risk in real time.

1.3 Objectives

1. Design a floating IoT-based water monitoring platform.
2. Measure water turbidity and optical scattering characteristics.

3. Utilize light scattering and multi-angle detection techniques.
4. Implement machine learning-based contamination classification.
5. Display contamination risk levels in real time.
6. Develop a low-cost prototype suitable for environmental monitoring.

1.4 Scope

The system estimates microplastic contamination risk rather than directly measuring exact microplastic concentration. The proposed design serves as an early warning and screening tool for water quality assessment.

Chapter 2: Literature Review

2.1 Microplastic Pollution

Microplastics have been detected in marine environments, freshwater systems, groundwater, and drinking water supplies. Their persistence in the environment has raised concerns regarding ecological and human health impacts.

2.2 Conventional Detection Techniques

Common laboratory methods include:

- FTIR Spectroscopy
- Raman Spectroscopy
- Microscopic Analysis
- Chemical Digestion Methods

Although highly accurate, these methods are expensive and impractical for real-time monitoring.

2.3 Optical Detection Methods

Optical sensing techniques utilize:

- Light transmission
- Light scattering
- Fluorescence response
- Turbidity measurements

Changes in optical properties can indicate the presence of suspended particles, including potential microplastic contaminants.

2.4 IoT-Based Environmental Monitoring

IoT systems provide continuous monitoring capabilities through:

- Embedded sensors
- Wireless communication

- Real-time data processing
- Cloud-based analysis

These technologies enable cost-effective environmental surveillance.

Chapter 3: Methodology

3.1 System Architecture

The proposed system consists of:

Hardware Components

- ESP32 Microcontroller
- Turbidity Sensor
- RGB LEDs
- UV LED
- Light Dependent Resistors (LDRs)
- OLED Display
- Relay Module
- DC Water Pump
- LM2596 Voltage Regulator
- Lithium-Ion Battery Pack

Software Components

- Arduino IDE
 - ESP32 Firmware
 - Machine Learning Classification Model
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3.2 Light Scattering and Multi-Angle Detection

The original detection concept uses optical scattering principles.

Working Principle

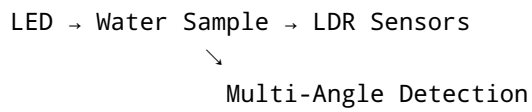
Light emitted by LEDs passes through the water sample.

When particles are present:

- Light intensity decreases.
- Scattering increases.
- Different angles receive different light levels.

Multiple LDR sensors positioned at different angles capture scattering information.

Detection Configuration



Changes in sensor readings indicate variations in particle concentration.

3.3 Turbidity-Based Risk Estimation

Due to practical implementation constraints, turbidity measurements are used as the primary indicator of contamination risk.

The turbidity sensor continuously measures water clarity.

Higher turbidity values generally indicate increased suspended particle concentration.

3.4 Machine Learning Classification

Sensor readings are used to classify contamination levels into:

- Very Low Risk
- Low Risk
- Medium Risk
- High Risk
- Very High Risk

Features include:

- Turbidity Value
- RGB Intensity Levels
- UV Response
- Multi-Angle Scattering Readings

Potential algorithms:

- Decision Tree
 - Random Forest
 - K-Nearest Neighbors
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Chapter 4: Results and Discussion

4.1 Experimental Setup

Water samples with varying particle concentrations were analyzed using the prototype.

Measurements collected:

- Turbidity
- Optical Intensity
- Scattering Patterns

4.2 Observations

Results indicated that increasing particle concentration produced:

- Higher turbidity values
- Increased light scattering
- Reduced transmitted light intensity

These changes allowed classification of contamination levels.

4.3 Risk Classification

Turbidity Range	Risk Level
0–100	Very Low Risk
101–250	Low Risk
251–500	Medium Risk
501–750	High Risk
Above 750	Very High Risk

4.4 Discussion

The prototype successfully demonstrated the feasibility of low-cost contamination risk assessment. Although the system cannot directly quantify microplastic concentration, it can provide valuable early-stage screening information.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

This project developed an IoT-based water quality monitoring system utilizing light scattering, multi-angle optical detection, and turbidity sensing. The system provides a practical method for estimating microplastic contamination risk using affordable hardware components.

The integration of machine learning enables automated classification of contamination levels, making the system suitable for real-time environmental monitoring applications.

5.2 Limitations

- Cannot directly measure exact microplastic concentration.
- Optical readings may be affected by non-plastic suspended particles.
- Environmental conditions can influence sensor performance.

5.3 Future Work

Future improvements may include:

- Advanced optical sensors
- Camera-based image analysis
- Deep learning models
- Cloud-based monitoring dashboards
- GPS-enabled contamination mapping
- Integration with laboratory validation methods

References

1. Research articles on microplastic contamination in aquatic environments.
2. Studies on optical sensing and light scattering techniques.
3. ESP32 IoT development documentation.
4. Environmental monitoring and machine learning research publications.
5. Water quality monitoring standards and guidelines.

Appendices

- Circuit Diagrams
- ESP32 Source Code
- Sensor Calibration Data
- Experimental Readings
- Machine Learning Dataset Samples